Exp 4	Design of Low Pass Filter using Windowing Techniques
Date:	**4.1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB**

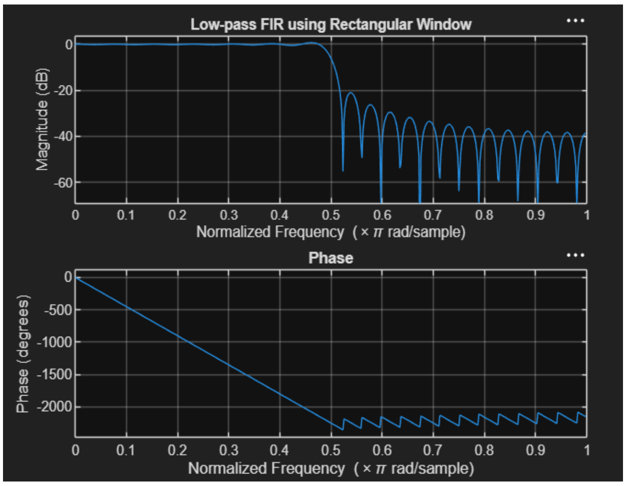
Code:

```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000;   % Passband cutoff (Hz)
fs = 4000;   % Sampling frequency (Hz)
N  = 50;     % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect');

%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');

%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
```

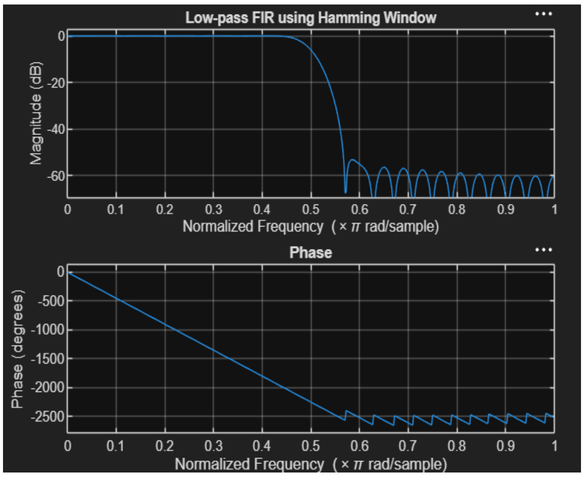
```
%% --- Plot Frequency Responses ---

figure;

freqz(b_rect,1);

title('Low-pass FIR using Rectangular Window');


figure;

freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window');


figure;

freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');
```

Output:

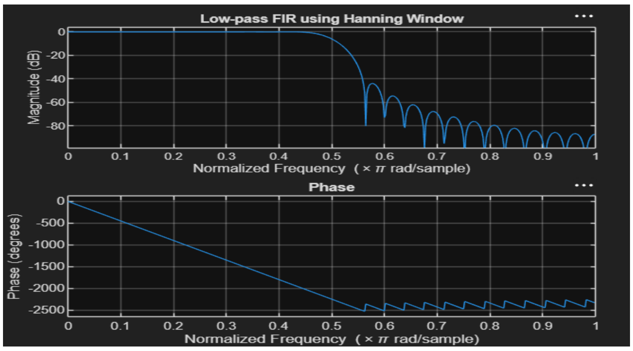
Objective 1:



Objective 2:



Objective 3:



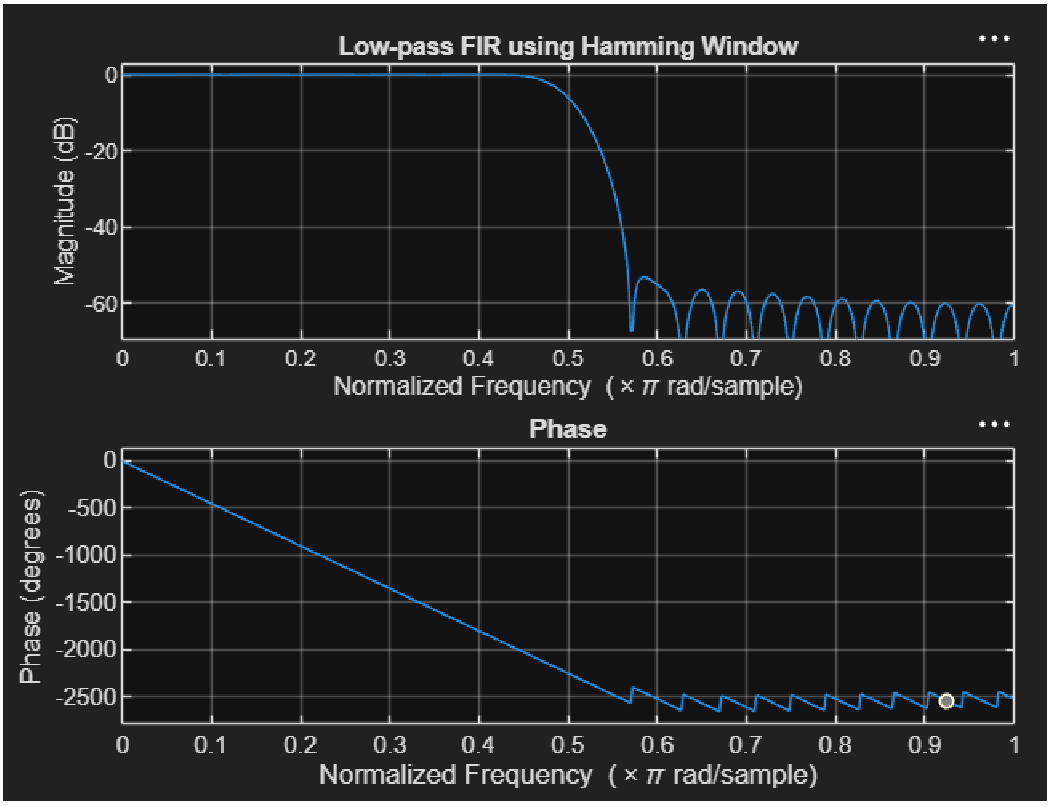
Exp 4	Design of Low Pass Filter using Windowing Techniques
Date:	4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique in MATLAB

Code:

```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000;   % Passband cutoff (Hz)
fs = 4000;   % Sampling frequency (Hz)
N  = 50;     % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');

%% --- Plot Frequency Response ---
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
```

Output:

Objective 1:



Exp 4	Design of Low Pass Filter using Windowing Techniques
Date:	**4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB**

Code:

```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000;   % Passband cutoff (Hz)
fs = 4000;   % Sampling frequency (Hz)
N  = 50;     % Filter order (even number preferred)
```

```
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');

%% --- Plot Frequency Response ---
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

Output:

Objective: